EWM104-BT60SP2&SP3 User Manual

AT Command Bluetooth Audio + Serial Data Transmission Module



Chengdu Ebyte Electronic Technology Co.,Ltd.

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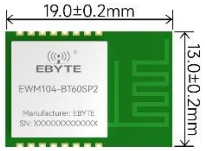
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1 Introduction

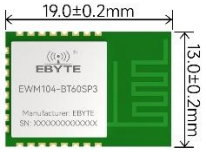
1.1 Brief Introduction

EWM104-BT60SP2/EWM104-BT60SP3 are new China produced, low-cost dual-mode Bluetooth slave modules for audio and data transmission launched by Chengdu EBYTE. Utilizing a domestic SOC integrated chip and adhering to Bluetooth 5.2 protocol specifications, they meet customers' low-cost Bluetooth audio requirements.

The EWM104-BT60SP2/EWM104-BT60SP3 supports the Bluetooth protocol stack including audio, voice, and SPP profiles. It provides a UART serial port, supports BLE/BR and EDR, and supports AT commands. Users can flexibly modify parameters such as serial port baud rate and device name as needed. The module can be customized for specific development requirements.



EWM104-BT60SP2



EWM104-BT60SP3

1.2 Features

- Bluetooth 5.2 protocol;
- Supports BLE and BR/EDR;
- Supports simultaneous Bluetooth audio playback and BLE data transmission;
- Maximum single-packet data transmission size of 63 bytes under BLE;
- Supports true Bluetooth wireless stereo;
- Supports AAC and SBC audio codecs;
- Supported Bluetooth profiles: A2DP, AVRCP, HFP, GATT, and SPP;
- Provides UART serial port with AT command support;
- Supports audio button control and microphone input;
- Connection range: 200m; communication range: up to 300m (in clear, open environments at maximum transmit power);
- EWM104-BT60SP2/EWM104-BT60SP3 module dimensions: 13×19mm;

1.3 Application

- Smart speakers
- Smart toys
- Massage chairs
- Smart home devices, etc.

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2 Operating Parameters

2.1 Specifications

Parameter Name	EWM104-BT60SP2	EWM104-BT60SP3	Note
Operating Frequency Band	2.4GHz		-
Communication Interface	UART Serial port		Supports AT commands
Serial Port Configuration	115200bps 8N1		-
Transmit Power	8dBm @MAX		Support Configuration
Receive Sensitivity	-95dBm		-
Connection Range	200m		-
Transmission Range	300m		-
Operating Voltage (V)	3.0~4.3		Typical value: 3.3V. Risk of burnout if exceeded.
Operating Temperature (°C)	-40~+85		Industrial-grade design
Operating Humidity (%)	10~90		Clear, open environment at maximum transmit power
Dimensions	13 * 19 mm		Clear, open environment at maximum transmit power
Antenna Interface	Postage Stamp Hole/PCB Antenna		Pinhole requires jumper resistor modification/equivalent impedance approx. 50Ω
Package Type	Surface Mount		-
Interface Type	Postage Stamp Hole		Pitch: 1.27mm

Average power consumption	EWM104-BT60SP2	EWM104-BT60SP3
Standby	1.8	1.8
BLE Connection (mA)	1.5	1.6
SPP Connection (mA)	2.6	2.5
BR/EDR Connection (mA)	2.6	2.5
BR/EDR Music Playback (mA)	4.1	4.3
BLE Data Pass-Through (mA)	1.5	1.8
SPP Data Pass-Through (mA)	2.8	2.8
BLE Data Pass-Through + BR/EDR Music Playback (mA)	4.1	4.4

Note: ① The EWM104-BT60SP2 adopts a 16-pin package, does not provide a button GPIO, and does not support OTA updates.

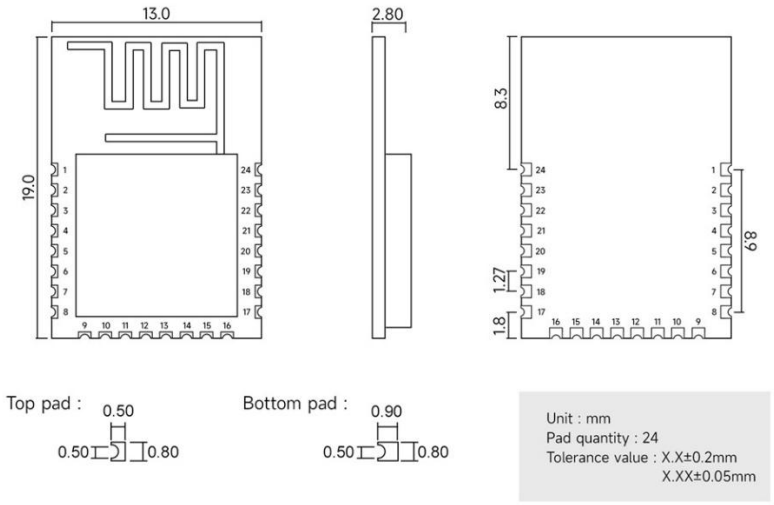
②The EWM104-BT60SP3 adopts a 24-pin package, provides a button GPIO, and supports OTA updates.

③When playing music, data can be transmitted simultaneously via BLE connection. If using SPP for data transmission, disconnect the music connection first.

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3 Mechanical Dimensions and Pin Definitions

3.1 EWM104-BT60SP3 Mechanical Dimensions and Pin Definitions

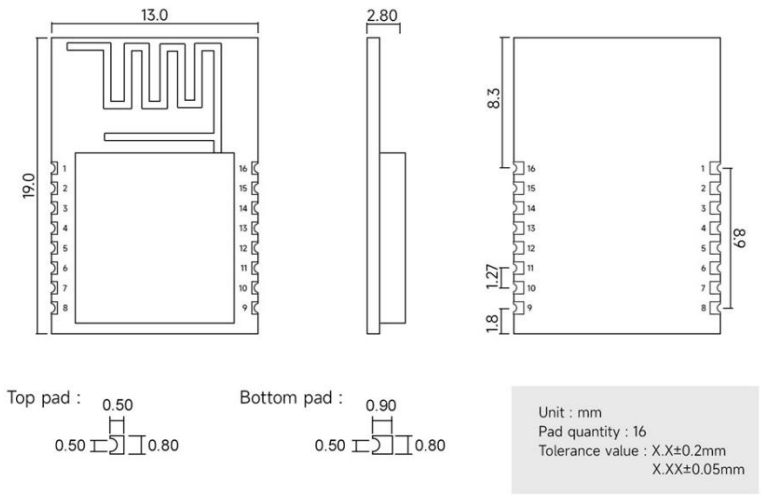


No	Name	Pin Function
1	AUDL_R	Audio Left Channel Positive Output
2	AUDL_N	Audio Left Channel Negative Output
3	GND	Ground
4	AUDR_N	Audio Right Channel Negative Output
5	AUDR_P	Audio Right Channel Positive Output
6	K1	Button GPIO: Active low. Single click: Play/Pause. Double click: Reject call. Long press: Disconnect Bluetooth.
7	MICBAIS	Microphone bias input. Leave floating when not in use.
8	MIC_P	Microphone channel positive input. Leave floating when not in use.
9	MIC_N	Microphone channel negative input. Leave floating when not in use.
10	GND	Ground
11	K2	Button GPIO2: Active low. Single click decreases volume; double click skips to previous track.
12	GND	Ground
13	MUTE	Power amplifier control pin. Outputs low level.
14	NC	Reserved pin. User does not need to concern themselves with this pin. Leave floating when not in use.
15	NC	Reserved pin. User does not need to concern themselves with this pin. Leave floating when not in use.
16	K3	Button GPIO3: Active low. Single click increases volume; double click skips to next track.
17	VBAT	Power Input Pin (Must be connected during operation, 3.0~4.3V)
18	VUSB	USB Charging Input (5V)
19	LED_A	Audio connection status pin, active high

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20	LED_L	BLE connection status pin, active high
21	RXD	Serial data RX input
22	TXD	Serial data TX output
23	GND	Ground
24	ANT	Stamp pad antenna (requires internal jumper resistor adjustment when in use)

3.2 EWM104-BT60SP2 Mechanical Dimensions and Pin Definitions



No	Name	Pin Function
1	AUDL_R	Audio Left Channel Positive Output
2	AUDL_N	Audio Left Channel Negative Output
3	GND	Ground
4	AUDR_N	Audio Right Channel Negative Output
5	AUDR_P	Audio Right Channel Positive Output
6	NC	Reserved pin; users need not concern themselves with this. Leave floating when not in use.
7	MICBAIS	Microphone Bias Input; leave floating when not in use.
8	MIC_P	Microphone Channel Positive Input; leave floating when not in use.
9	VBAT	Power Input Pin (Must be connected during power supply operation; 3.0–4.3V)
10	VUSB	USB Charging Input (5V)
11	LED_A	Audio Connection Status Pin, Active High
12	LED_L	BLE Connection Status Pin, Active High
13	RXD	Serial Data RX Input
14	TXD	Serial Data TX Output
15	GND	Ground
16	ANT	Stamp Pad Antenna (Internal jumper resistor adjustment required when in use)

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4 Quick Command Test

4.1 Audio Usage

After powering on the module, you can scan for Bluetooth devices using your phone. Search for the E104-BT60_BR/EDR broadcast and connect to it. Audio playback can then begin immediately.

4.2 SPP Usage

After powering on the module, use the SPP debugging software on your phone to scan and connect to the E104-BT60_BR/EDR broadcast. The module will then enter configuration mode upon receiving the input “+++”. In configuration mode, send the command AT+TRANSMODE=2 to set it to SPP transmission mode. Finally, send AT+EXIT to exit configuration mode, enabling transparent data transmission.

4.3 BLE Usage

After powering on the module, use a mobile BLE debugging app to scan and connect to the E104-BT60_BLE broadcast. Once connected, open the FFF1 notification channel to enable transparent data transmission from the module to the mobile device. Data can be transmitted from the mobile device to the module via the FFF2 channel and ultimately printed through the serial port.

5 Instructions and Factory Settings

5.1 Command Format

The module supports parameter reading and writing. The default baud rate for the module's serial port is 115200 bps, 8N1 format.

Command Format	Description	Example
AT+XXX	Run Command	AT+RESET
AT+XXX=?	Query Command	AT+TRANSMODE=?
AT+XXX=YYYY	Set Command	AT+TRANSMODE=1

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5.2 Return Value Description

AT Command Return Value	Explanation
AT_OK	Correct
+ERROR：Error code	+ERROR：Error code

Error code	Error Description
-1	Length error
-2	Hyperparameter range
-3	No parameters
-4	Not supported
-5	Flash write error
-6	Format error

5.3 AT Command List

Command Type	Instruction	Description
Basic Commands	AT	AT Test Commands
	AT+RESTORE	Restore Factory Settings
	+++	Enter AT Mode
	AT+TRANSMODE	Data Pass-Through Channel Commands
	AT+VER	Read Version Number
	AT+RESET	Reboot
	AT+EXIT	Exit AT Mode
	AT+BAUD	Baud Rate Commands
	AT+POWER	Transmit Power Commands
BT command	AT+NAME	BR/EDR Name Commands
	AT+MAC	BR/EDR MAC Commands
	AT+BLE_NAME	BLE Name Commands
	AT+BLE_MAC	BLE MAC Commands
	AT+BLEDISCON	Disconnect BLE Connection
	AT+BTDISCON	Disconnect BR/EDR Connection
	AT+PLAYPAUSE	Play/Pause Commands
	AT+NEXT	Next Track Commands
	AT+PREV	Previous Track Commands
	AT+VOLP	Volume Up
	AT+VOLM	Volume Down

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5.4 AT Command Reference

5.4.1 AT Test Command

Command	Response	Instruction
AT	+OK\r\n	**Test Command** **Send: AT** **Return: +OK\r\n**

5.4.2 AT+RESTORE Command

Command	Response	Instruction
AT+RESTORE	+OK\r\n	**Restore Factory Settings** **Send: AT+RESTORE** **Return: +OK\r\n**

5.4.3 Enter AT commands

Command	Response	Instruction
+++	AT_MODE	**Enter AT Mode** **Send: +++** **Return: AT_MODE**

5.4.4 AT+TRANSMODE command

Command	Response	Instruction
AT+TRANSMODE	TRANSMODE:BLE\r\n	**Query/Set Data Pass-Through Channel** **Send: AT+TRANSMODE=<CMD>\r\n** **Return: TRANSMODE: <CMD>\r\n** **<Cmd> Parameter Description** **?: Read** **BLE: BLE channel** **SPP: SPP channel**

5.4.5 AT+VER command

Command	Response	Instruction
AT+VER	VER:xxxx-x-xx\r\n	**Read version number** **Send: AT+VER**

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		Return: VER:xxxx-x-xx\r\n Read version number Send: AT+VER Return: VER:xxxx-x-xx\r\n

5.4.6 AT+RESET command

Command	Response	Instruction
AT+RESET	+OK\r\n	**Reset Module** **Send: AT+RESET** **Return: +OK\r\n AT+RESET**

5.4.7 AT+EXIT command

Command	Response	Instruction
AT+EXIT	+OK\r\n	**Exit AT Mode** **Send: AT+EXIT** **Return: +OK\r\n**

5.4.8 AT+BAUD command

Command	Response	Instruction
AT+BAUD=<cmd>	BOND:<cmd>\r\n	**Query/Set Baud Rate** **<Cmd> Parameter Description** **?: Read** 0:4800 1:9600 2:19200 3:38400 4:57600 5:115200 (Default) 6:230400 7:460800 8:921600 9:2000000

5.4.9 AT+POWER command

Command	Response	Instruction
AT+POWER=<cmd>	POWER:<cmd>\r\n	**Query/Set Transmission Power** **cmd Parameter Description** **?: Read** 0: 8dBm

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		1: 3dBm 2: 0dBm 3: -4dBm

5.4.10 AT+NAME command

Command	Response	Instruction
AT+NAME=<cmd>	BR/EDR_NAME:<cmd>\r\n	**Query/Set BR/EDR Name** **CMD Parameter Description** **?: Read** **String within 1-25 bytes** **Example: AT+NAME=E104-BT60_BR/EDR**

5.4.11 AT+MAC command

Command	Response	Instruction
AT+MAC=?	MAC:xxxxxxxxxxxx\r\n	**MAC Read Command**

5.4.12 AT+BLE_NAME command

Command	Response	Instruction
AT+BLE_NAME=<cmd>	BLE_NAME:<cmd>\r\n	**Query/Set BLE Name** **CMD Parameter Description** **?: Read** **String within 1-25 bytes** **Example: AT+BLE_NAME=E104-BT60_BLE**

5.4.13 AT+BLE_MAC command

Command	Response	Instruction
AT+BLE_MAC=<cmd>	BLE_MAC=<cmd>\r\n	**Query/Set BLE_MAC** **CMD Parameter Description** **?: Read** **String within 12 bytes** **Example: AT+BLE_MAC=112233445566**

5.4.14 AT+BLEDISCON command

Command	Response	Instruction
AT+BLEDISCON	+OK\r\n	Disconnect BLE connection

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		Send: AT+BLEDISCON Return: +OK\r\n

5.4.15 AT+BTDISCON command

Command	Response	Instruction
AT+BTDISCON	+OK\r\n	**Disconnect Bluetooth connection** **Send: AT+BTDISCON** **Return: +OK\r\n**

5.4.16 AT+PLAYPAUSE Command

Command	Response	Instruction
AT+PLAYPAUSE	+OK\r\n	**Pause/Play** **Send: AT+PLAYPAUSE** **Return: +OK\r\n**

5.4.17 AT+NEXT command

Command	Response	Instruction
AT+NEXT	+OK\r\n	**Next Track** **Send: AT+NEXT** **Return: +OK\r\n**

5.4.18 AT+PREV command

Command	Response	Instruction
AT+PREV	+OK\r\n	**Previous Track** **Send: AT+PREV** **Return: +OK\r\n**

5.4.19 AT+VOLP command

Command	Response	Instruction
AT+VOLP	+OK\r\n	**Volume Up** **Send: AT+VOLP** **Return: +OK\r\n**

5.4.20 AT+VOLM command

Command	Response	Instruction

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AT+VOLM	+OK\r\n	**Volume Down** **Send: AT+VOLM** **Return: +OK\r\n**

6 Hardware design

- It is recommended to use a DC regulated power supply to power this module, the power supply ripple factor is as small as possible, and the module needs to be reliably grounded;
- Please pay attention to the correct connection of the positive and negative terminals of the power supply, such as reverse connection may cause permanent damage to the module;
- Please check the power supply to ensure that it is between the recommended supply voltages, if it exceeds the maximum value it may cause permanent damage to the module;
- Please check the power supply stability, the voltage should not fluctuate significantly and frequently;
- In the design of power supply circuit for the module, it is often recommended to retain more than 30% of the margin, there is the whole machine is conducive to long-term stable work;
- Module should be as far away as possible from the power supply, transformers, high-frequency alignments and other electromagnetic interference in the larger part;
- High-frequency digital alignment, high-frequency analog alignment, power supply alignment must be avoided below the module, if you really need to go through the module below, assuming that the module is welded in the Top Layer, the Top Layer in the contact part of the module to lay the ground copper (all paved with copper and a good ground), it must be close to the digital part of the module and alignment in the Bottom Layer;
- Assuming that the module is soldered or placed in the Top Layer, it is also wrong to randomly route the module in the Bottom Layer or any other layer, which will affect the spuriousness of the module as well as the reception sensitivity to varying degrees;
- Assuming that there is a large electromagnetic interference around the module device will also greatly affect the performance of the module, according to the intensity of the interference is recommended to stay away from the module, if the situation permits you can do appropriate isolation and shielding;
- Assume that there is a large electromagnetic interference around the module alignment (high-frequency digital, high-frequency analog, power supply alignment) will also greatly affect the performance of the module, according to the intensity of the interference is recommended to be appropriate away from the module, if the situation permits you can do appropriate isolation and shielding;
- Communication line if you use 5V level, must be connected in series with 1k-5.1k resistor (not recommended, there is still a risk of damage);
- Try to stay away from some TTL protocols where the physical layer is also 2.4GHz, e.g. USB3.0;
- The antenna mounting structure has a big impact on the module performance, make sure the antenna is exposed and preferably vertically upward;
- When the module is installed inside the chassis, use a good quality antenna extension cable to extend the antenna to the outside of the chassis;
- The antenna must not be installed inside the metal shell, which will lead to a great weakening of the transmission distance.

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7 Common problems

7.1 Power-on module shows no response

- Communication distance will decrease when there are direct line-of-sight obstructions;
- Temperature, humidity, and co-channel interference can increase packet loss rates;
- Ground absorption and reflection of radio waves result in poor testing performance near the ground;
- Seawater strongly absorbs radio waves, leading to poor testing results near the coast;
- Metal objects near the antenna or housing it in a metal enclosure cause severe signal attenuation;
- Incorrect power register settings or excessively high air interface rate (higher rates reduce communication distance);
- Power supply voltage below recommended levels at room temperature reduces transmit power;
- Using antennas poorly matched to the module or of inferior quality.

7.2 Modules are fragile

- Please check the power supply to ensure that it is between the recommended supply voltages, if it exceeds the maximum value it will cause permanent damage to the module;
- Please check the power supply stability, the voltage can not be substantial frequent fluctuations;
- Please ensure that the installation and use process anti-static operation, high-frequency device electrostatic sensitivity;
- Please ensure that the installation and use of the process of humidity should not be too high, part of the components for humidity-sensitive devices;
- If there is no special demand is not recommended to be used at too high or too low a temperature.

7.3 Short transmission distance/becoming shorter

- Module transmission performance degradation (low transmit power, reduced receive sensitivity), module damage, return for factory repair;
- Antenna loosening, corrosion, or damage, tighten antenna connections, replace antenna;
- New interference sources, prioritize channel change, adjust/upgrade antenna;

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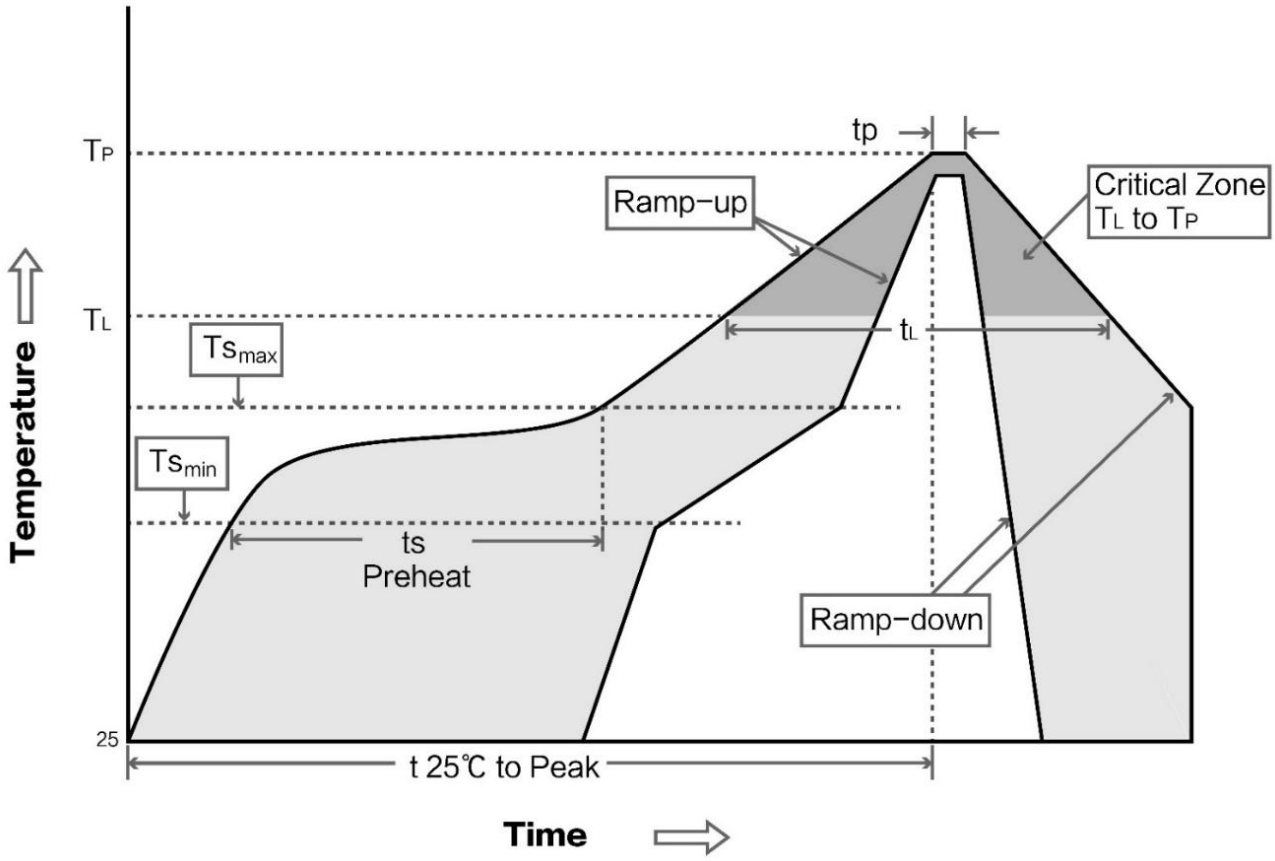
8 Welding instructions

8.1 Reflow temperature

Profile Feature	Curve Characteristics	Sn-Pb Assembly	Pb-Free Assembly
Solder Paste	Solder Paste	Sn63/Pb37	Sn96.5/Ag3/Cu0.5
Preheat Temperature min （Tsmin）	Minimum Preheat Temperature	100℃	150℃
Preheat temperature max (Tsmax)	Maximum Preheat Temperature	150℃	200℃
Preheat Time (Tsmin to Tsmax)(ts)	Preheat Time	60-120 sec	60-120 sec
Average ramp-up rate(Tsmax to Tp)	Average Rise Rate	3℃/second max	3℃/second max
Liquidous Temperature (TL)	Liquid phase temperature	183℃	217℃
Time（tL）Maintained Above（TL）	Time above liquid phase line	60-90 sec	30-90 sec
Peak temperature（Tp）	Peak temperature	220-235℃	230-250℃
Aveage ramp-down rate（Tp to Tsmax）	Average rate of descent	6℃/second max	6℃/second max
Time 25℃ to peak temperature	Time from 25℃ to peak temperature	6 minutes max	8 minutes max

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8.2 Reflow temperature



Revision history

Version	Date	Description	Issued by
1.0	2025-9-22	Initial version	Lei
1.1	2025-11-12	Update the mechanical dimension drawing	Lei
1.2	2026-06-03	Modify pad dimensions	Lei

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